

Lab - Extracting Thumbnail Images From a Suspects Machine

Overview

In this lesson, you will learn how to extract thumbnail images from the thumbcache_*.db and iconcache_*.db database files. Law enforcement agencies have used thumbnail cache files to prove that a file of interest was stored on a Windows system's hard drive even if deleted. This is because when a user deletes a file, its thumbnail remains in the cached file.

Thumbcache files are databases that contain thumbnail images of content on your system. When you hover your mouse over an image in a folder, a preview of that image is generated. Windows will store this thumbnail in a thumbcache database for later retrieval (should you ever need to preview the image again).

The caching of thumbnail images speeds up the display of images as these smaller images do not need to be recalculated every time the user views the image folder. Thus, it takes less system resources (CPU time and memory) to retrieve an already generated thumbnail as opposed to generating a new one each time the image is viewed.

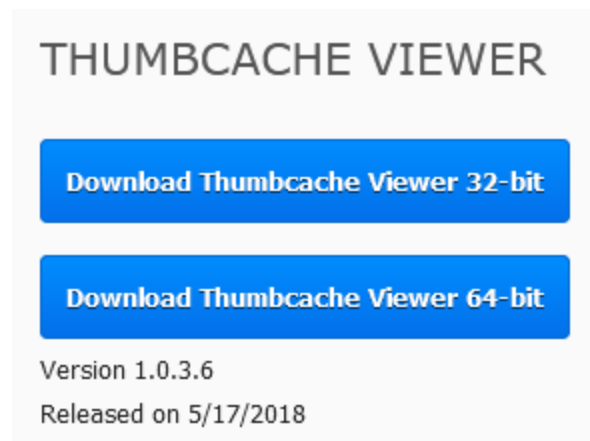
Lab Requirements

- One installation of Windows 7 or 10
- One downloaded copy of Thumbcache Viewer

Begin the lab!

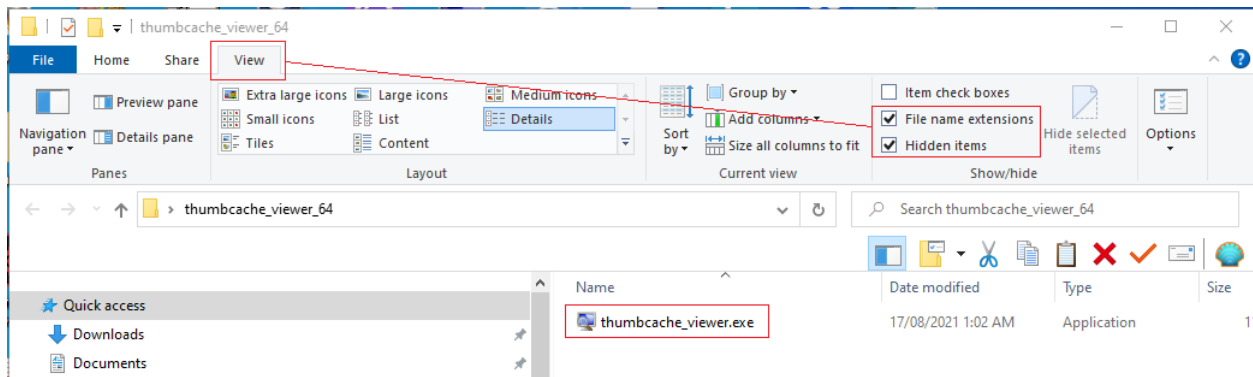
Download Thumbcache Viewer using the following URL.

<https://thumbcacheviewer.github.io/>

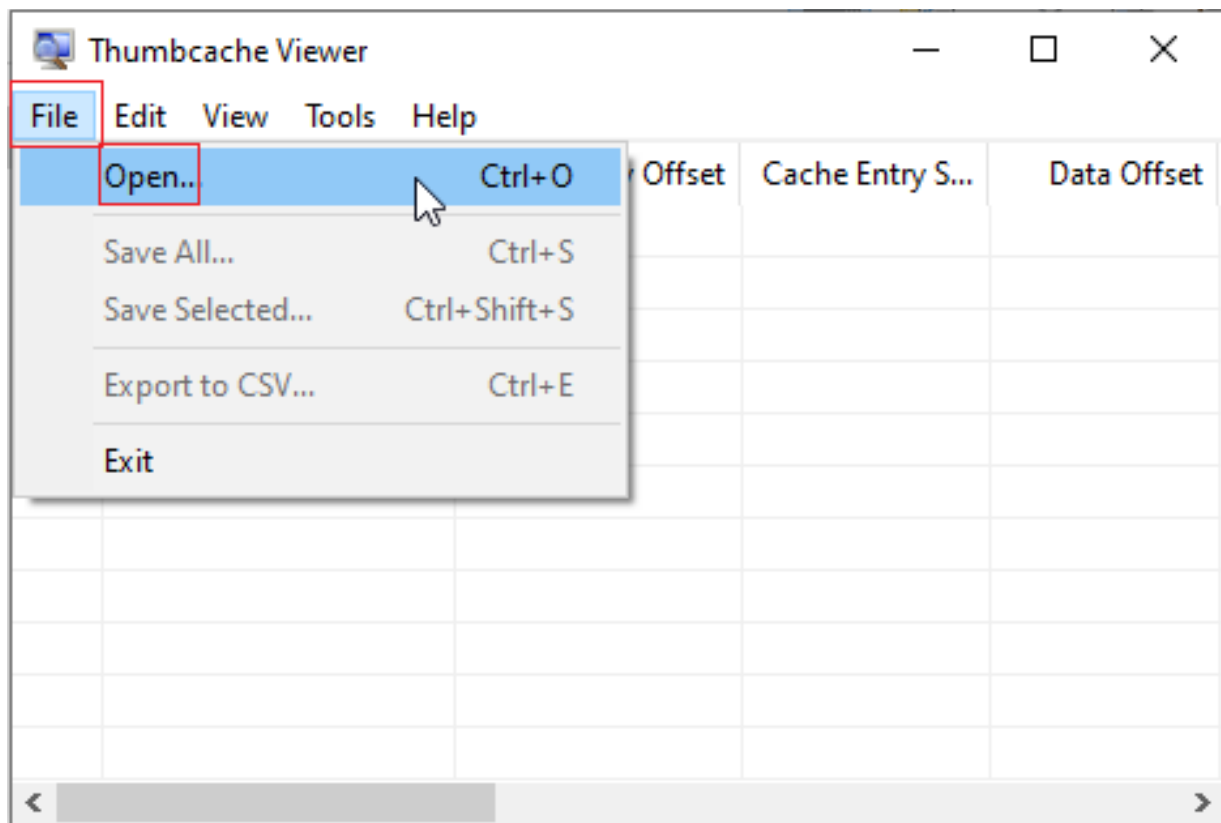


Extract the contents of the zip file. Then, save the contents to your desktop.

Open the extracted folder. You are now inside of Windows File Explorer. From the taskbar, click on the View tab. Check the boxes to show File name extensions and hidden items.

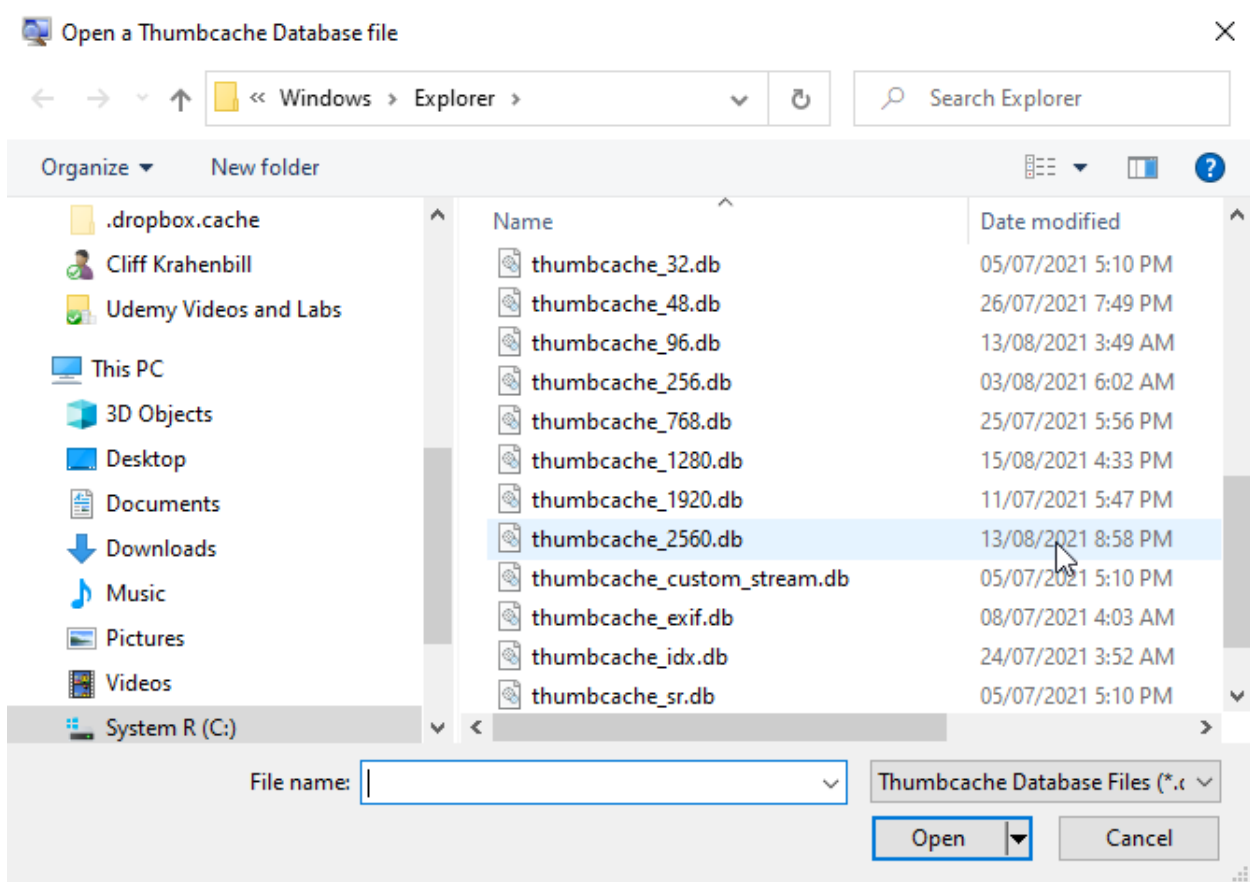


Launch the Thumbcache Viewer executable. No installation is required. Click on File and the Open.

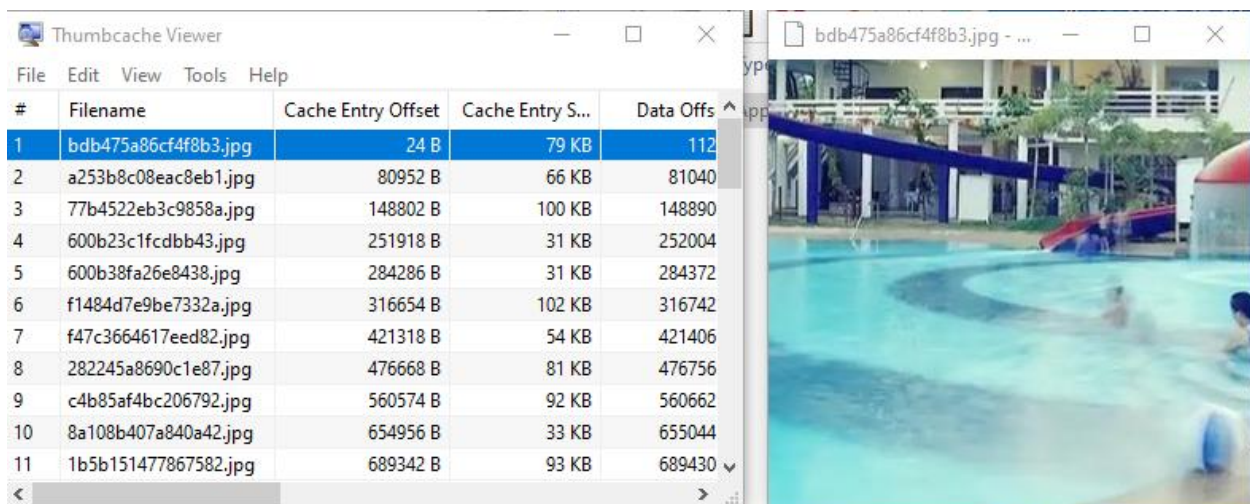


Each user has a thumbnail cache located in their local profile. Use the following path to see the thumbnail cache for the user currently logged on to the machine.

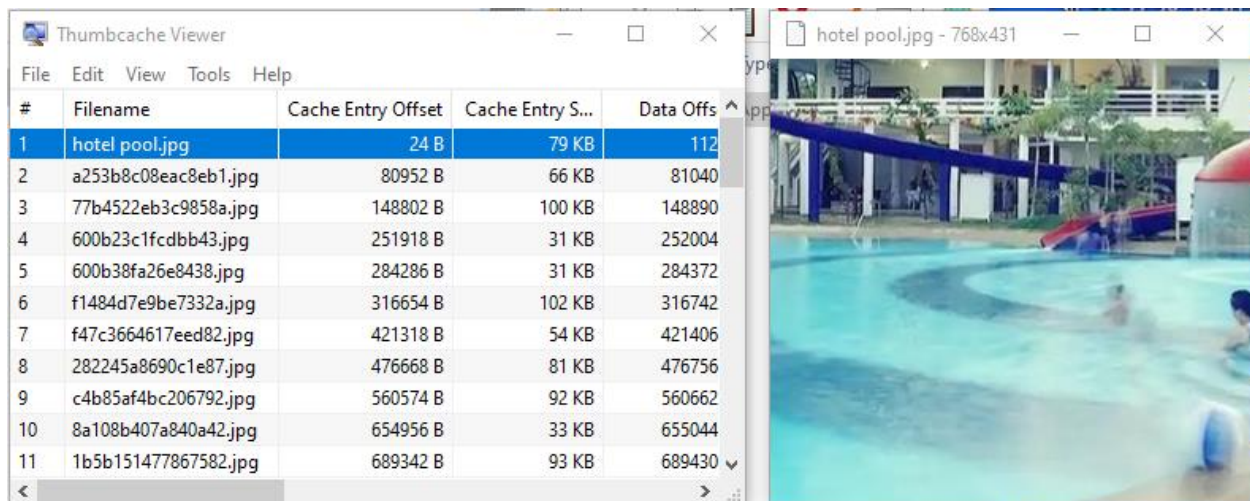
`C:\Users\<UserName>\AppData\Local\Microsoft\Windows\Explorer`



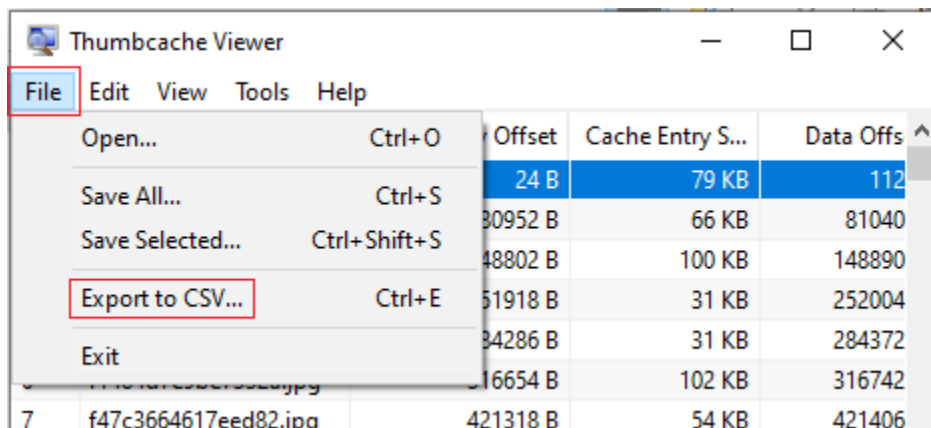
X2 click to view the contents of a thumbcache database. When you select an item for the list, the thumbnail for the image will pop out to the viewer's right.



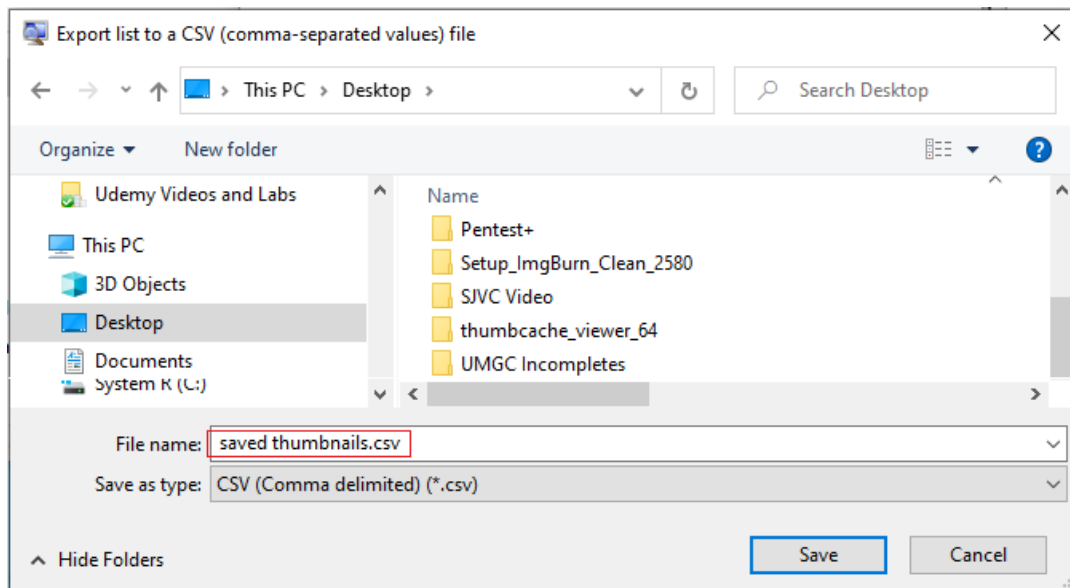
You can rename a file by clicking on the file's name and typing in the new name.



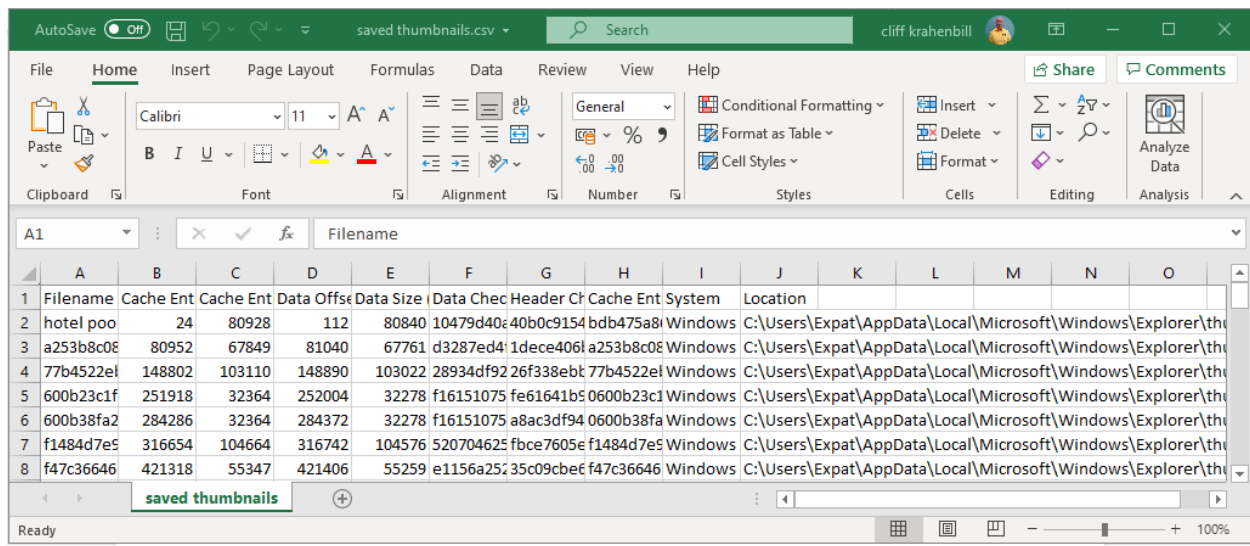
To save the thumbnail database to an Excel spreadsheet, click on File from the taskbar and select Export to CSV from the context menu.



Select a save location and give the CSV file a user-friendly name.



Open your saved CSV file.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Filename	Cache Ent	Cache Ent	Data Offs	Data Size	Data Chec	Header Ch	Cache Ent	System	Location					
2	hotel poo	24	80928	112	80840	10479d40e40b0c9154	bdb475a8	Windows	C:\Users\Expat\AppData\Local\Microsoft\Windows\Explorer\the						
3	a253b8c0e	80952	67849	81040	67761	d3287ed4	1dece406	a253b8c0e	Windows	C:\Users\Expat\AppData\Local\Microsoft\Windows\Explorer\the					
4	77b4522e	148802	103110	148890	103022	28934df9226f338eb	77b4522e	Windows	C:\Users\Expat\AppData\Local\Microsoft\Windows\Explorer\the						
5	600b23c1f	251918	32364	252004	32278	f16151075	fe61641b5	0600b23c1	Windows	C:\Users\Expat\AppData\Local\Microsoft\Windows\Explorer\the					
6	600b38fa2	284286	32364	284372	32278	f16151075	a8ac3df94	0600b38fa	Windows	C:\Users\Expat\AppData\Local\Microsoft\Windows\Explorer\the					
7	f1484d7e5	316654	104664	316742	104576	520704625	fbce7605e	f1484d7e5	Windows	C:\Users\Expat\AppData\Local\Microsoft\Windows\Explorer\the					
8	f47c36646	421318	55347	421406	55259	e1156a25435c09cbe	f47c36646	Windows	C:\Users\Expat\AppData\Local\Microsoft\Windows\Explorer\the						

End of the lab!